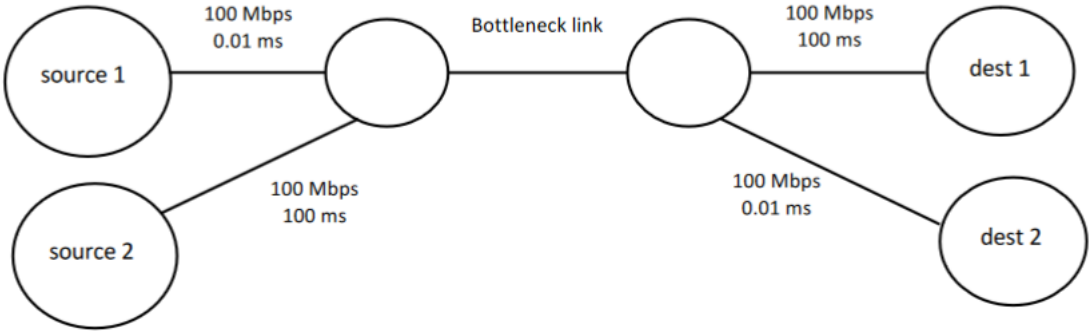
Félix Arnold

ECE 6110 Lab Assignment 2
Congestion Control

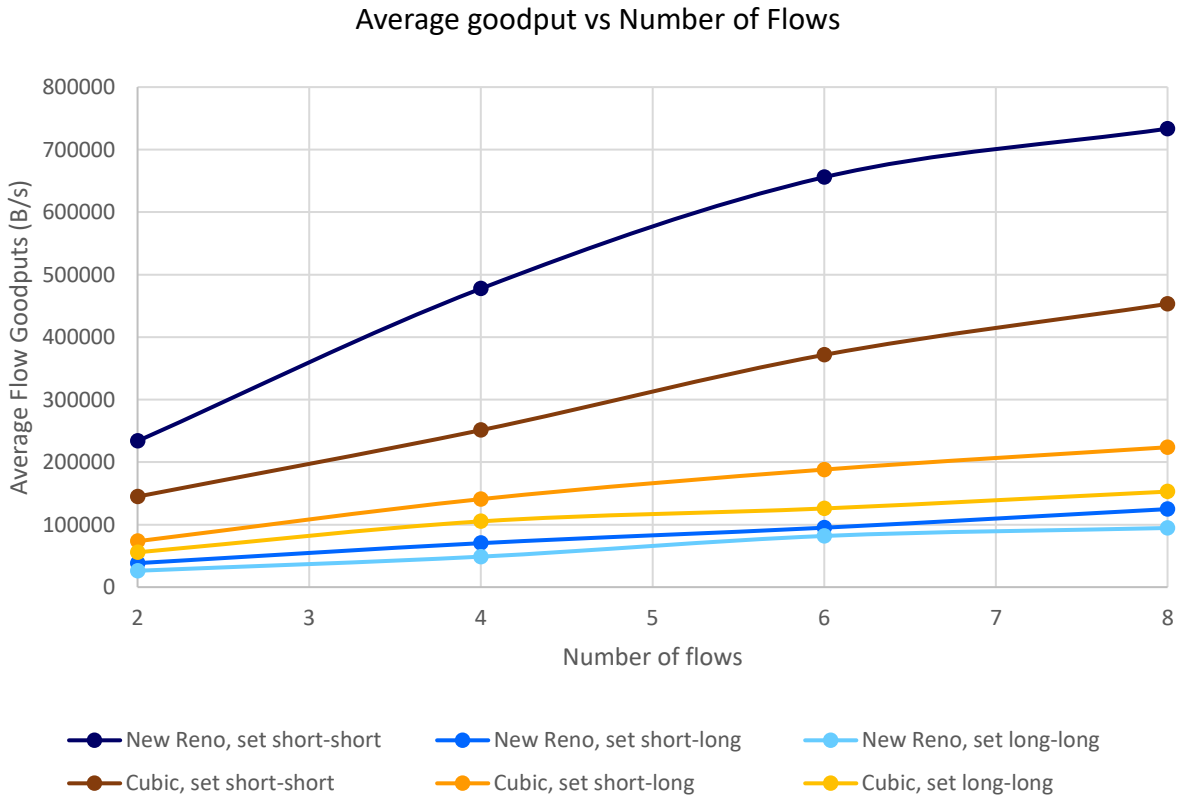
Part 2: Topology with a Bottleneck Link and Heterogeneous Flows

Now, we want to simulate the following topology:



For a bottleneck link data rate of 10 Mbps, a bottleneck link delay a 10 ms, and a byte error rate of 0.00001, we can plot average flow goodputs for the three sets of flows versus the number of flows per source.

We get the following graph using a bash script:



We observe that for all configurations and sets, the average flow goodput increase with the number of flows. We can also observe that flows with lower RTT have a higher goodput than flows with longer ones, since they can forward more packets in a same time.

The TCP Cubic achieve higher goodputs for all number of flows and set, except for the case with two short delay links. As such the TCP Cubic is fairer across all flows than New Reno because, using it, the goodputs are much more similar between the three different sets. These results are consistent with part 1 where Cubic also outperformed New Reno most of the time and has more stable results.